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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

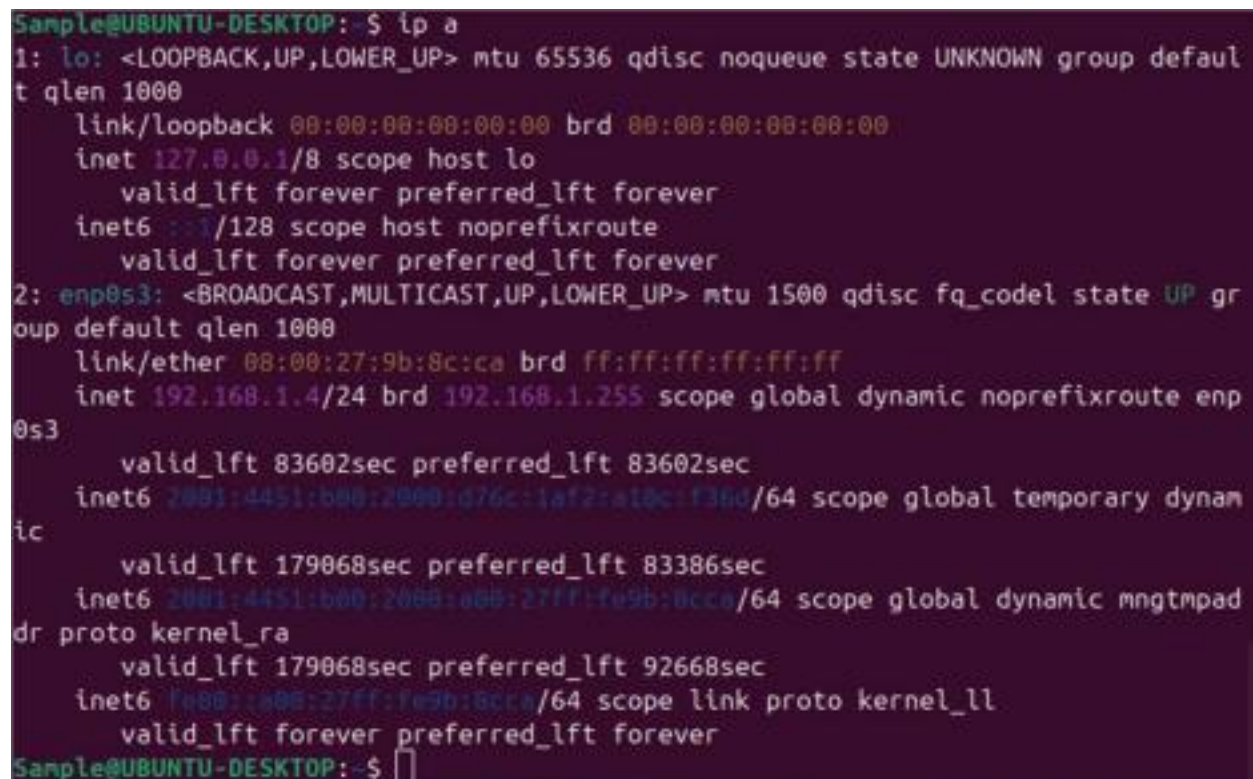
Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

A terminal window with a dark purple background and green text. The prompt is 'Sample@UBUNTU-DESKTOP:~\$'. The command 'ip a' has been executed, showing details for the loopback interface 'lo' (127.0.0.1) and the ethernet interface 'enp0s3' (192.168.1.4). The output for 'enp0s3' includes a MAC address and several IPv6 addresses. The prompt returns at the bottom.

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtproto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Student ID: 2025301449

Course/Year/Section: BSIT3B

Date Submitted: 02/09/2026

Instructor: [Johneros P. Puyo](#)

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ✓ Create and configure a Virtual Machine with Ubuntu Server
- ✓ Configure Bridged Network Adapter for network access
- ✓ Determine and verify the server's IP address
- ✓ Install OpenSSH Server and enable the service
- ✓ Establish successful SSH connection from host machine
- ✓ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Linux
Virtualization Software (VirtualBox/VMware)	Oracle Virtual Box
Ubuntu Server Version	ubuntu-26.04-live-server-amd64
VM Name	cja-server
RAM Allocated	8000MB
Storage Allocated	100GB
Network Mode	NatNetwork
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	8000 MB

Hard Disk	100 GB
Network Adapter	NatNetwork

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: 10.0.2.4

Network Interface: enp0s3

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	10.0.2.4
Username	aranas
SSH Port	22

Creating Vm server



Create Virtual Machine

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name: 

Folder:

ISO Image:

Edition:

Type: 

Version:

☐ Skip Unattended Installation

 Detected OS type: Ubuntu (64-bit). This OS type can be installed unattendedly. The install will start after this wizard is closed.

Create Virtual Machine

?

×



Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Additional Options

Username: ✓

Password: ✓

Repeat Password: ✓

Product Key:

Hostname: ✓

Domain Name:

☐ Install in Background

☐ Guest Additions

Guest Additions ISO:

Help

Back

Next

Cancel

Create Virtual Machine

?

×



Hardware

You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count. Enabling EFI is also possible.

Base Memory:

Processors:

4 MB

16384 MB

1 CPU

12 CPUs

☐ Enable EFI (special OSes only)

Help

Back

Next

Cancel

Create Virtual Machine

Virtual Hard disk

If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.

☒ Create a Virtual Hard Disk Now

Disk Size:

100.00 GB

4.00 MB
2.00 TB

☐ Pre-allocate Full Size

☐ Use an Existing Virtual Hard Disk File

Almendral-VM.vdi (Normal, 100.00 GB)

☐ Do Not Add a Virtual Hard Disk

Help

Back

Next

Cancel

Create Virtual Machine

Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.

Unattended Install

Username	aranas-server
Product Key	false
Hostname/Domain Name	aranas-server.myquest.virtualbox.org
Install in Background	false
Install Guest Additions	false

Hardware

Base Memory	8000
Processor(s)	4
EFI Enable	false

Disk

Disk Size	100.00 GB
Pre-allocate Full Size	false

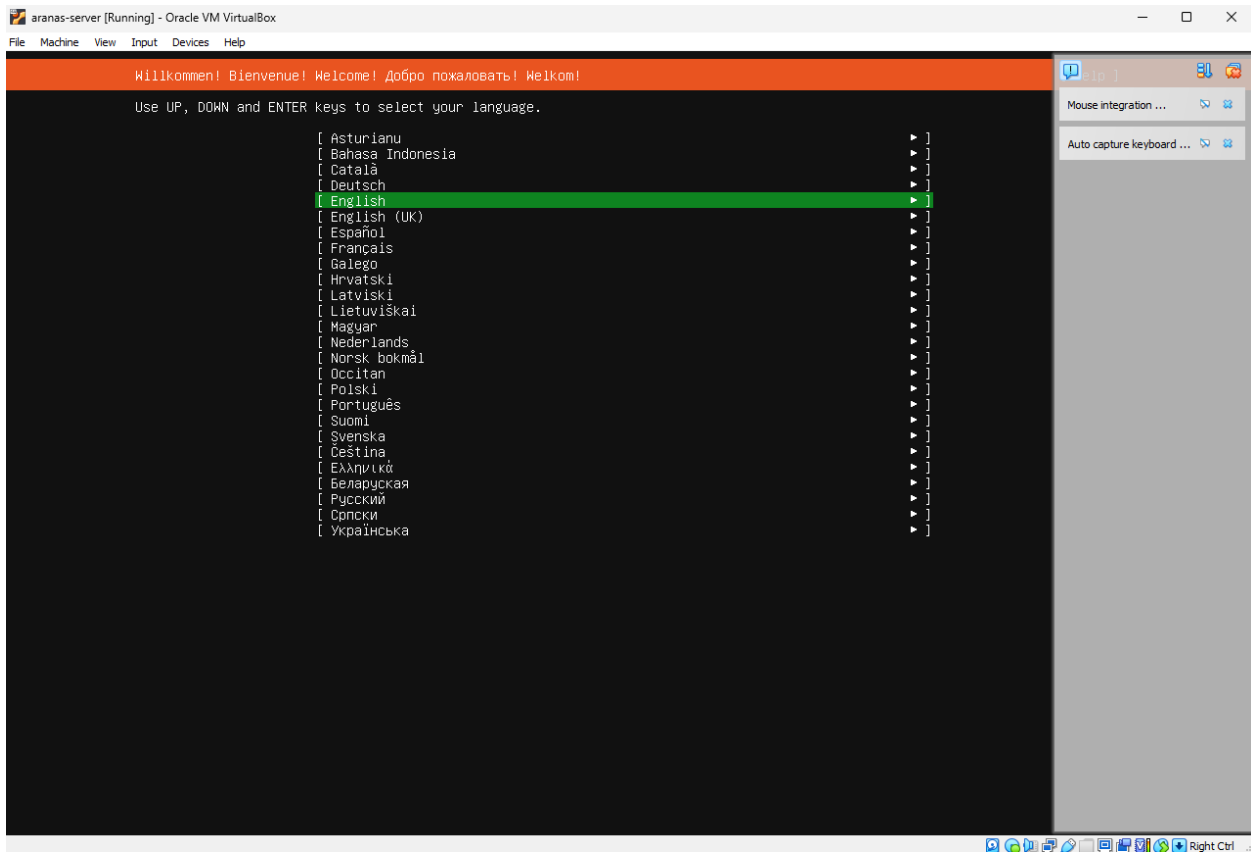
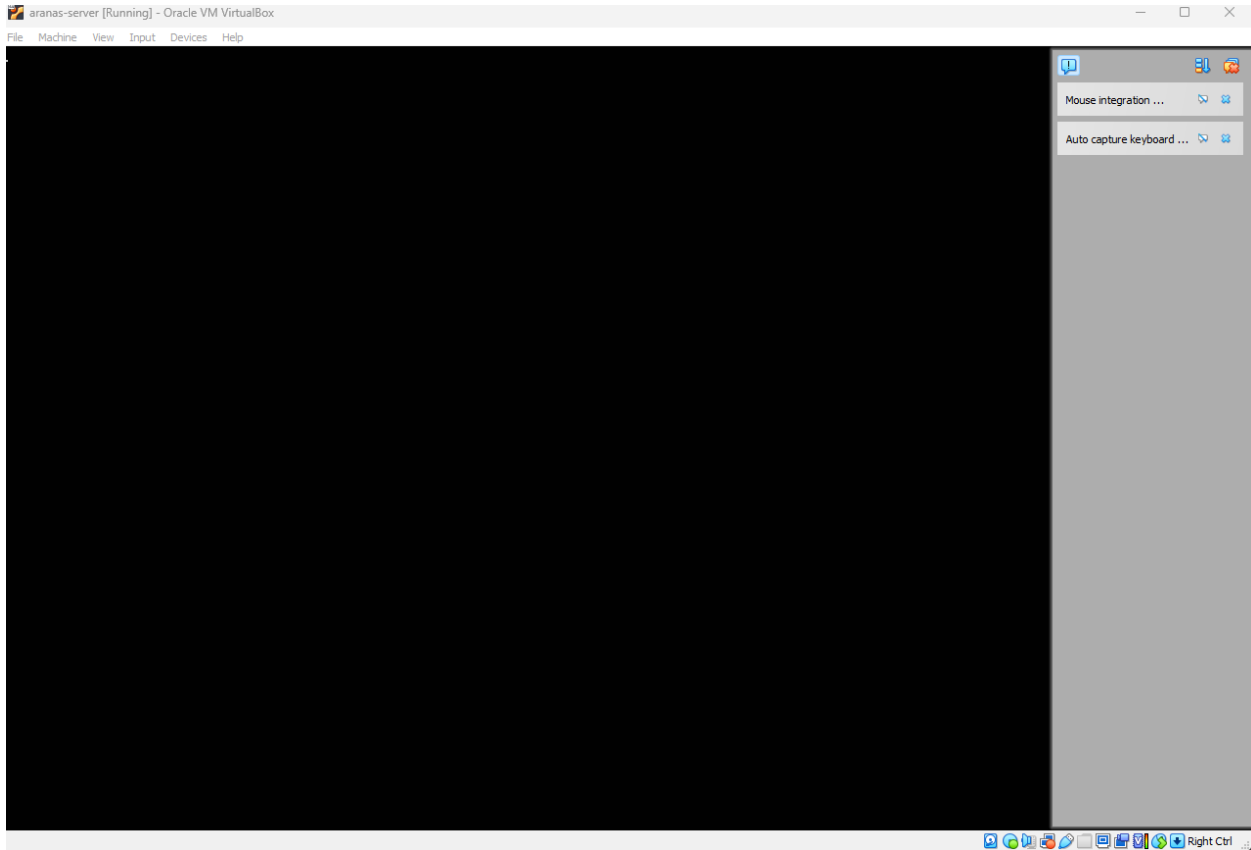
Help

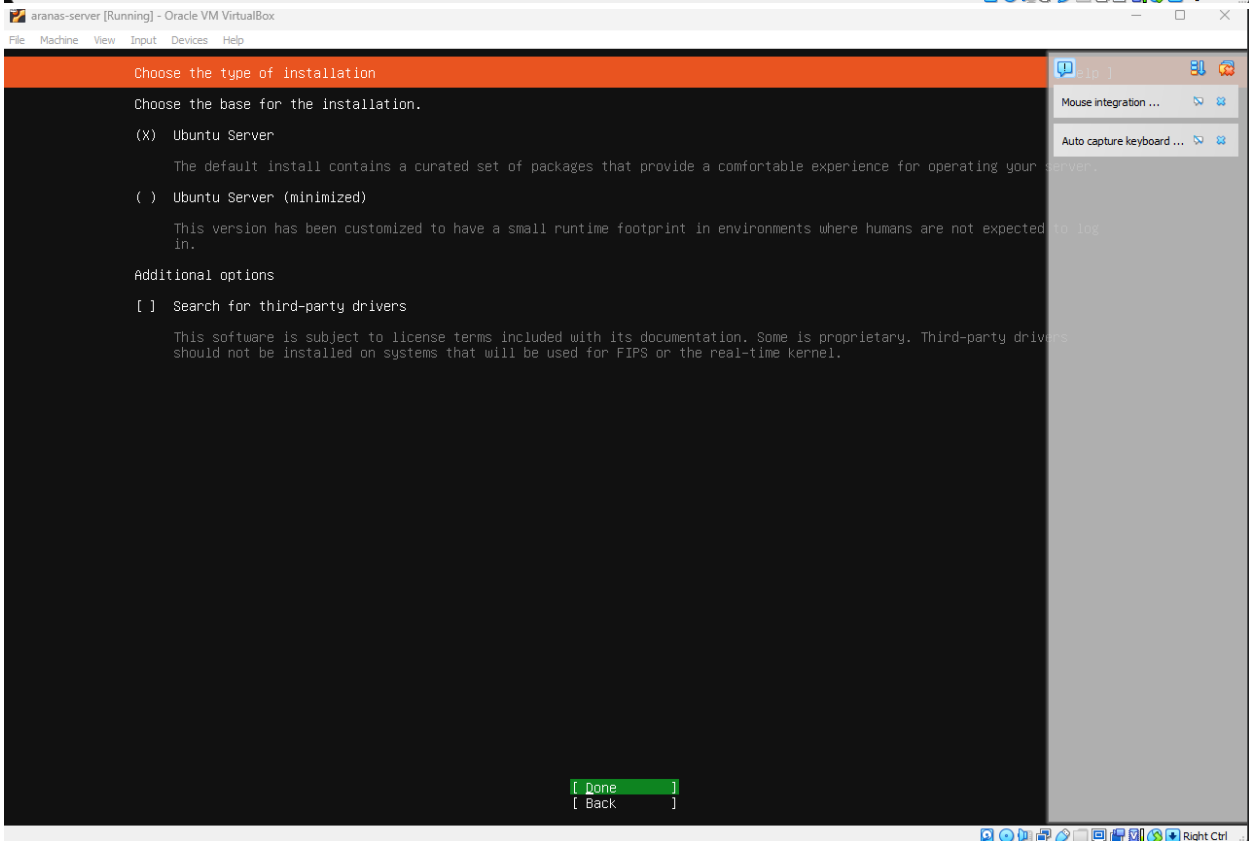
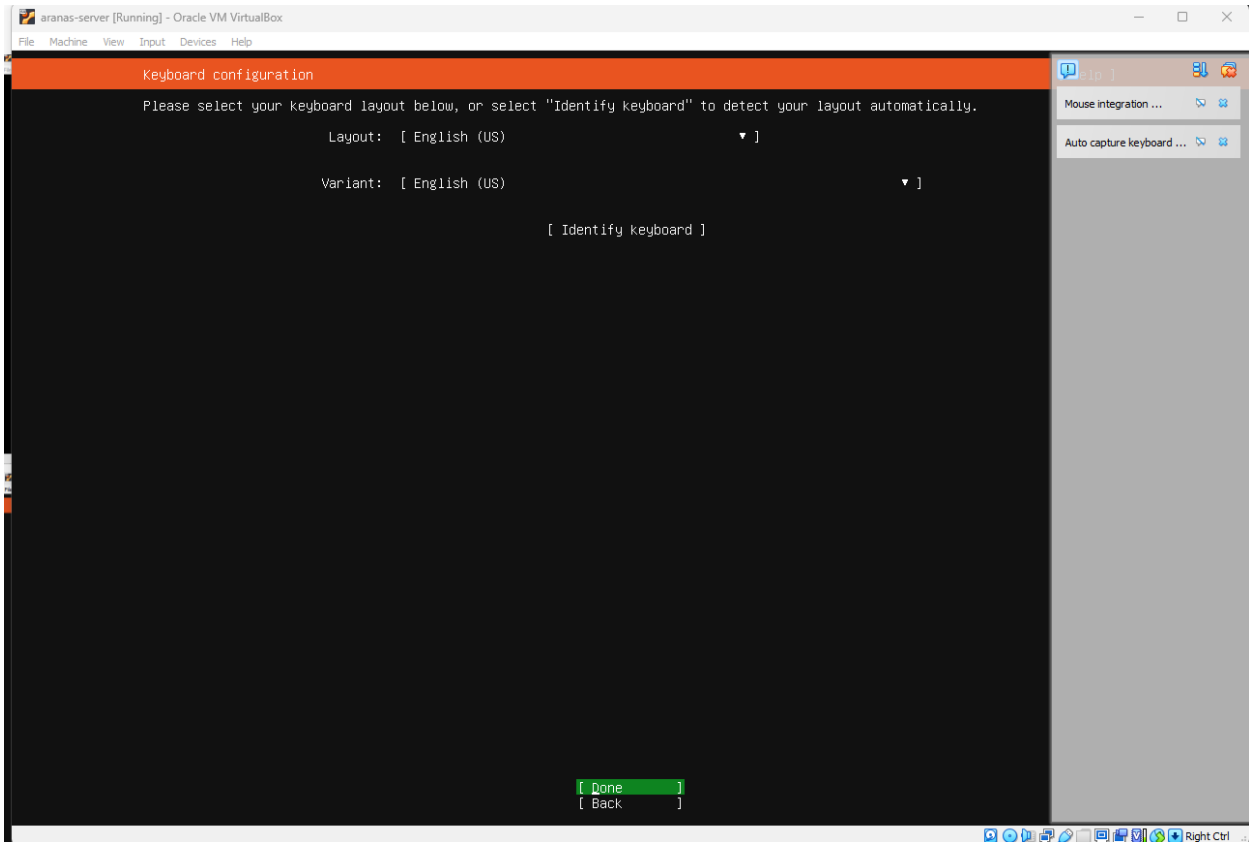
Back

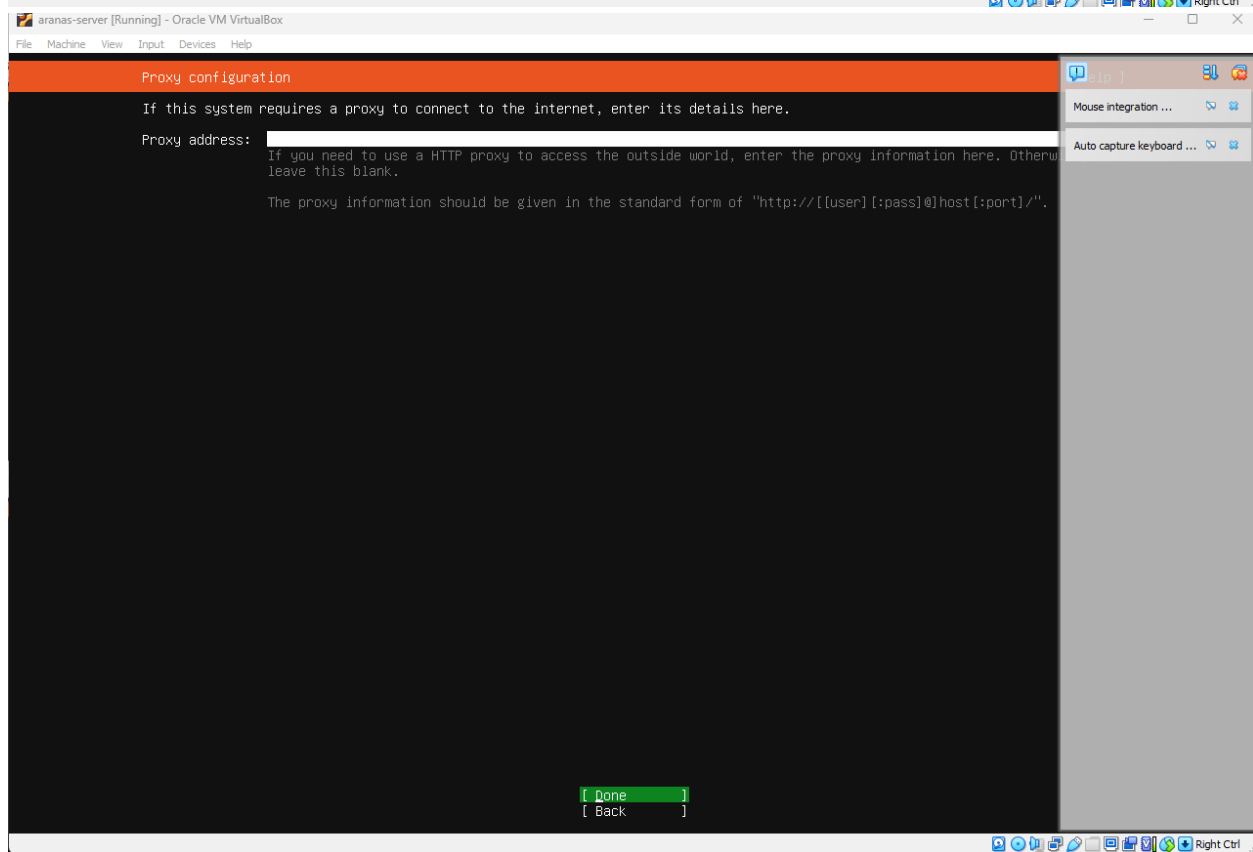
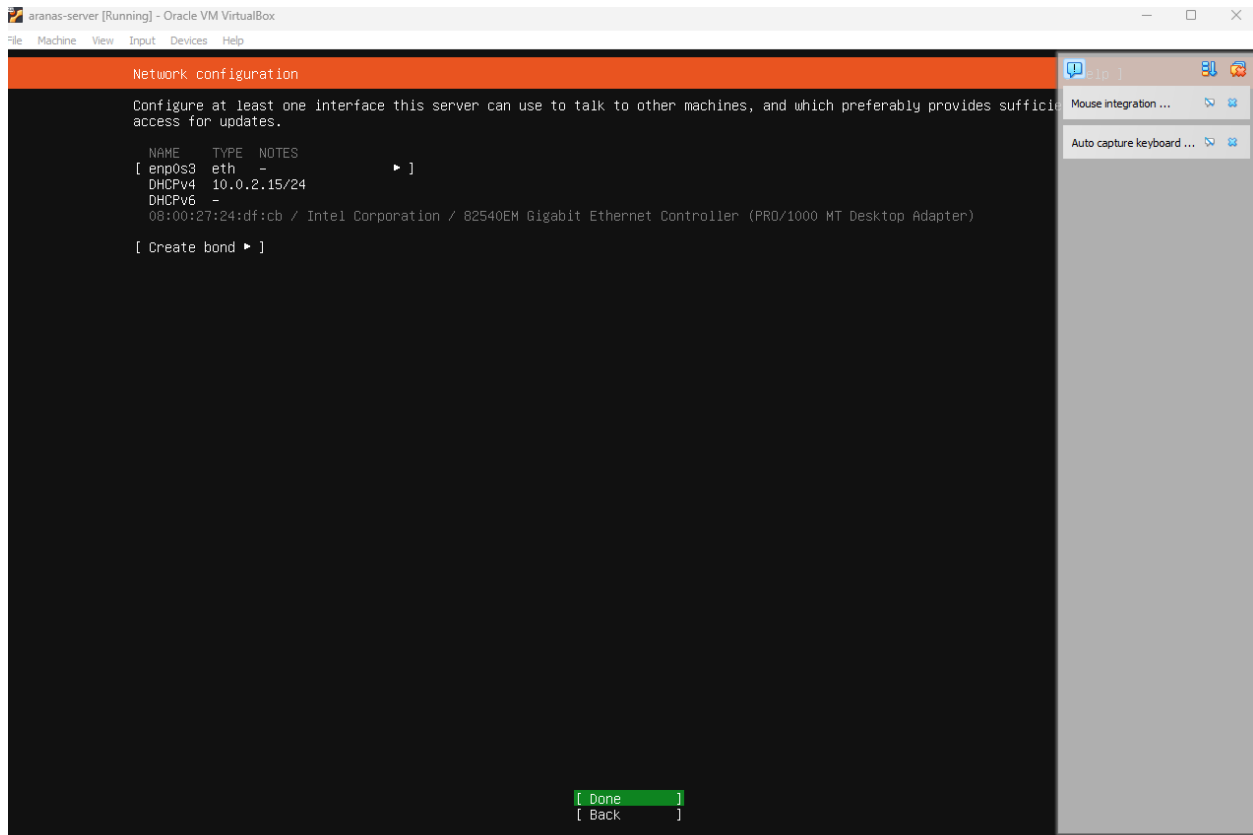
Finish

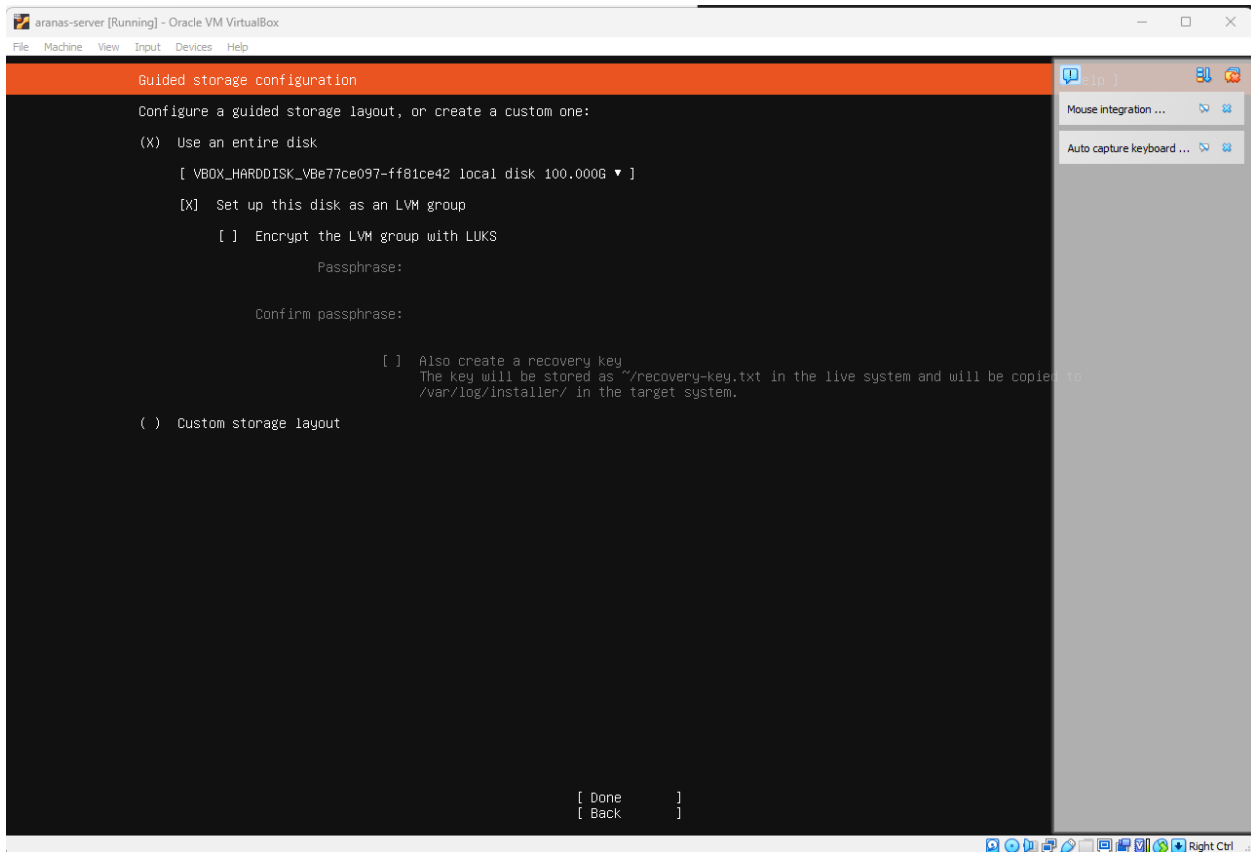
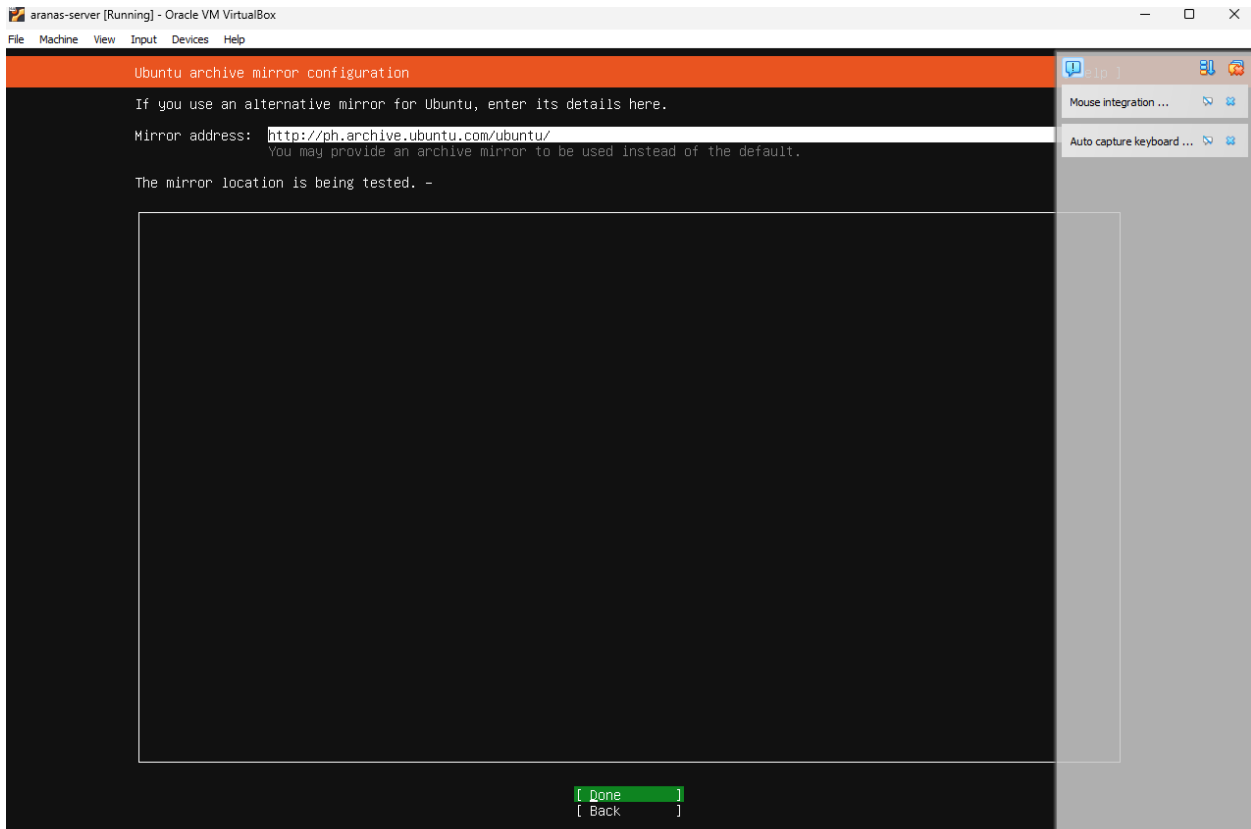
Cancel

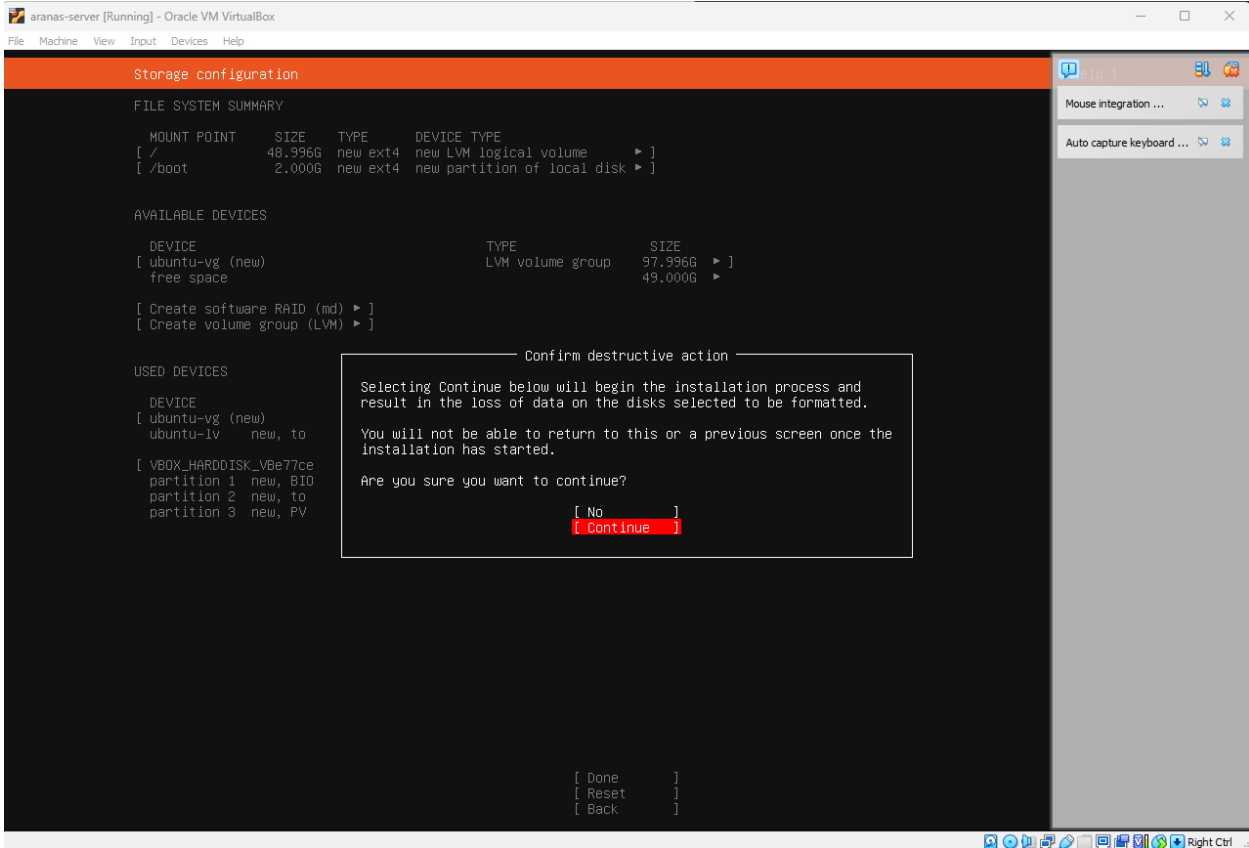
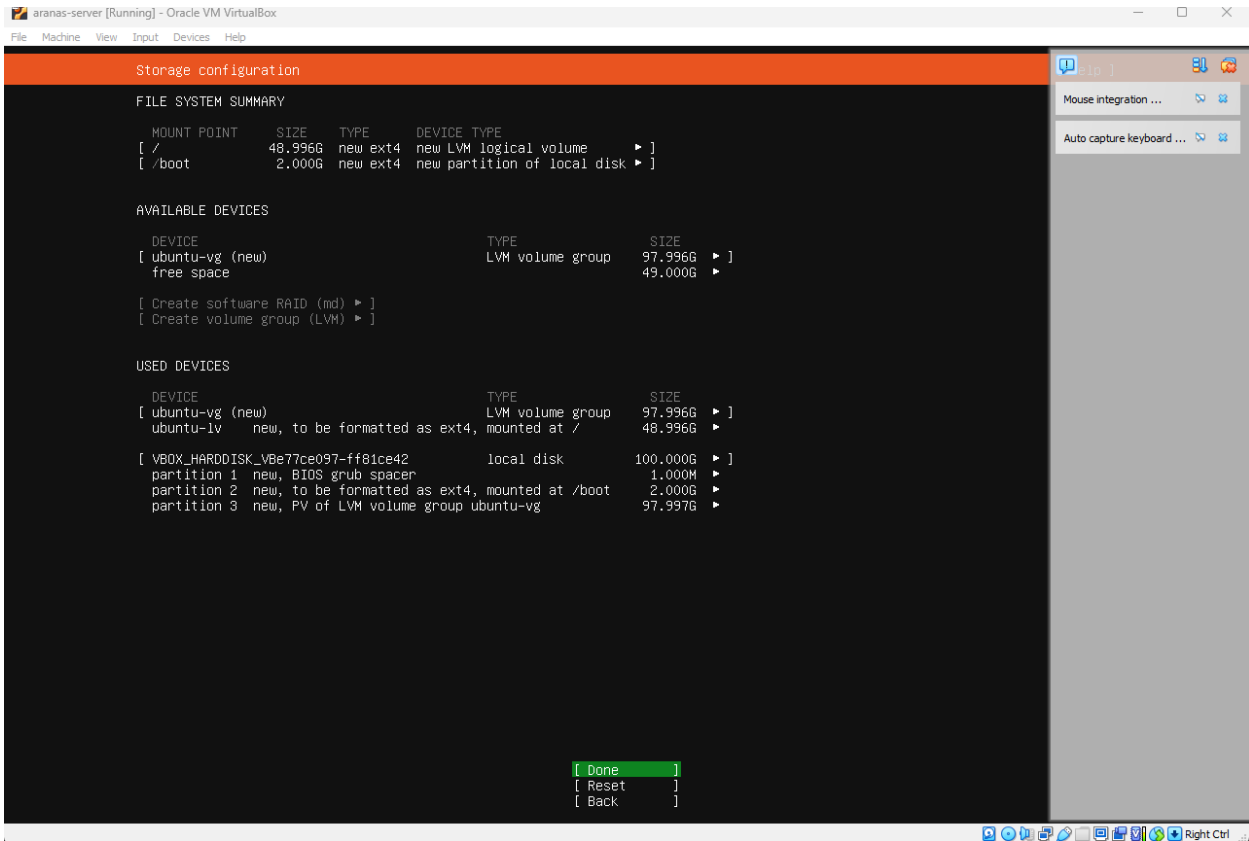
INSTALLATION PROCESS

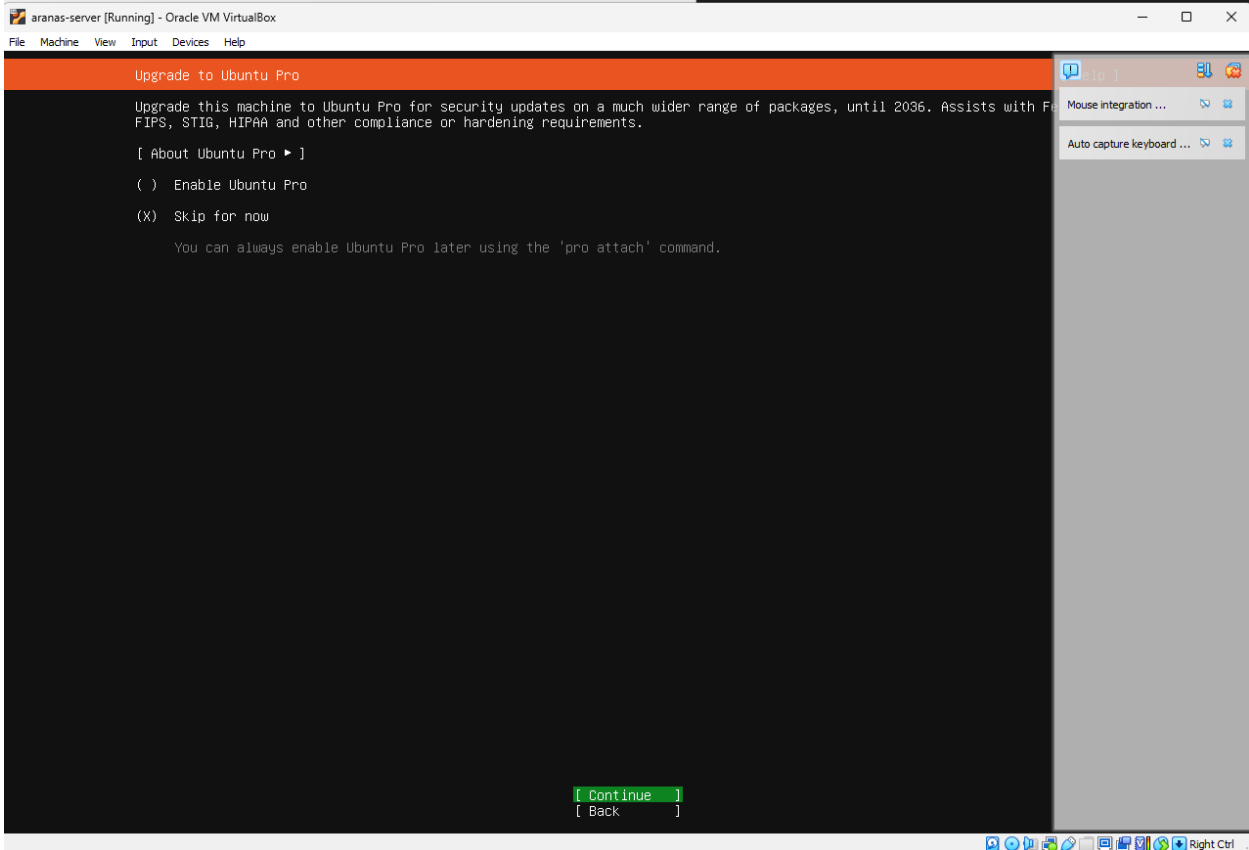
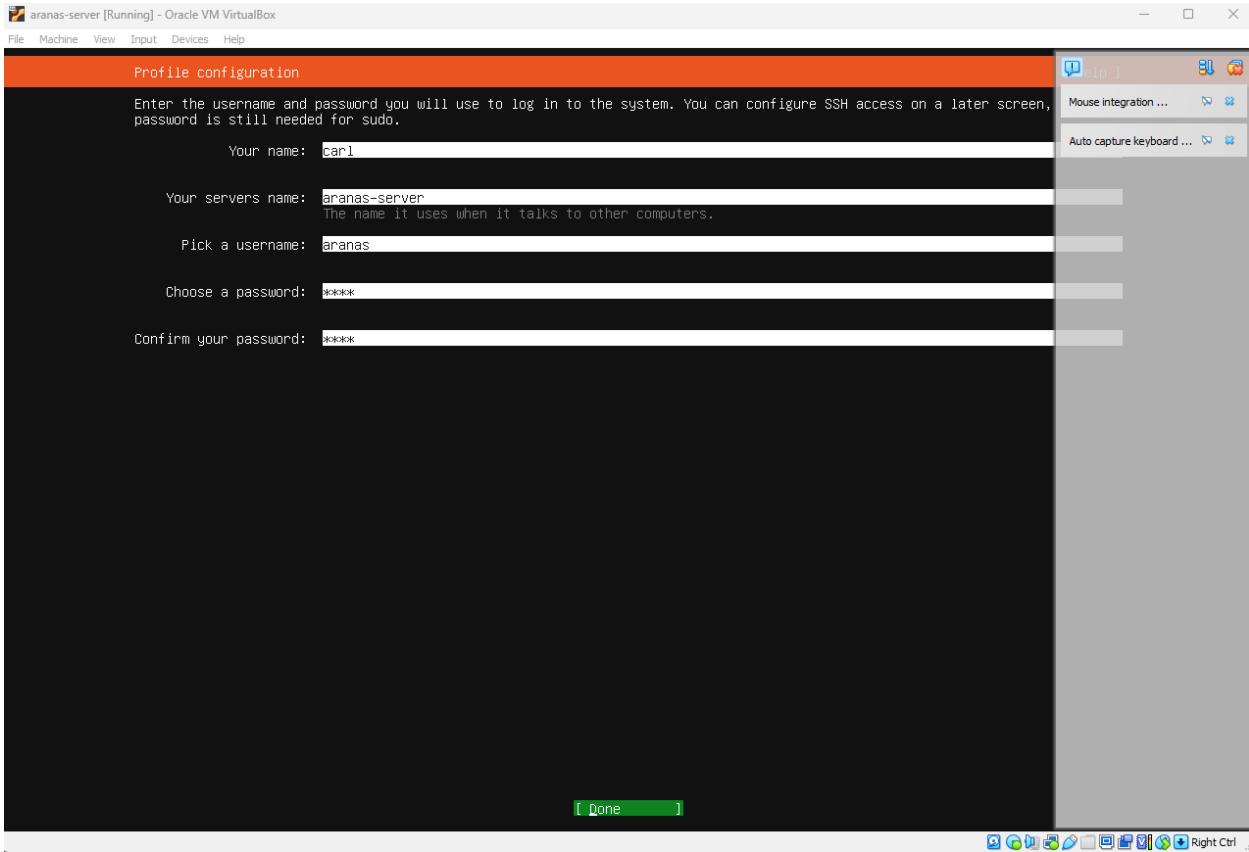


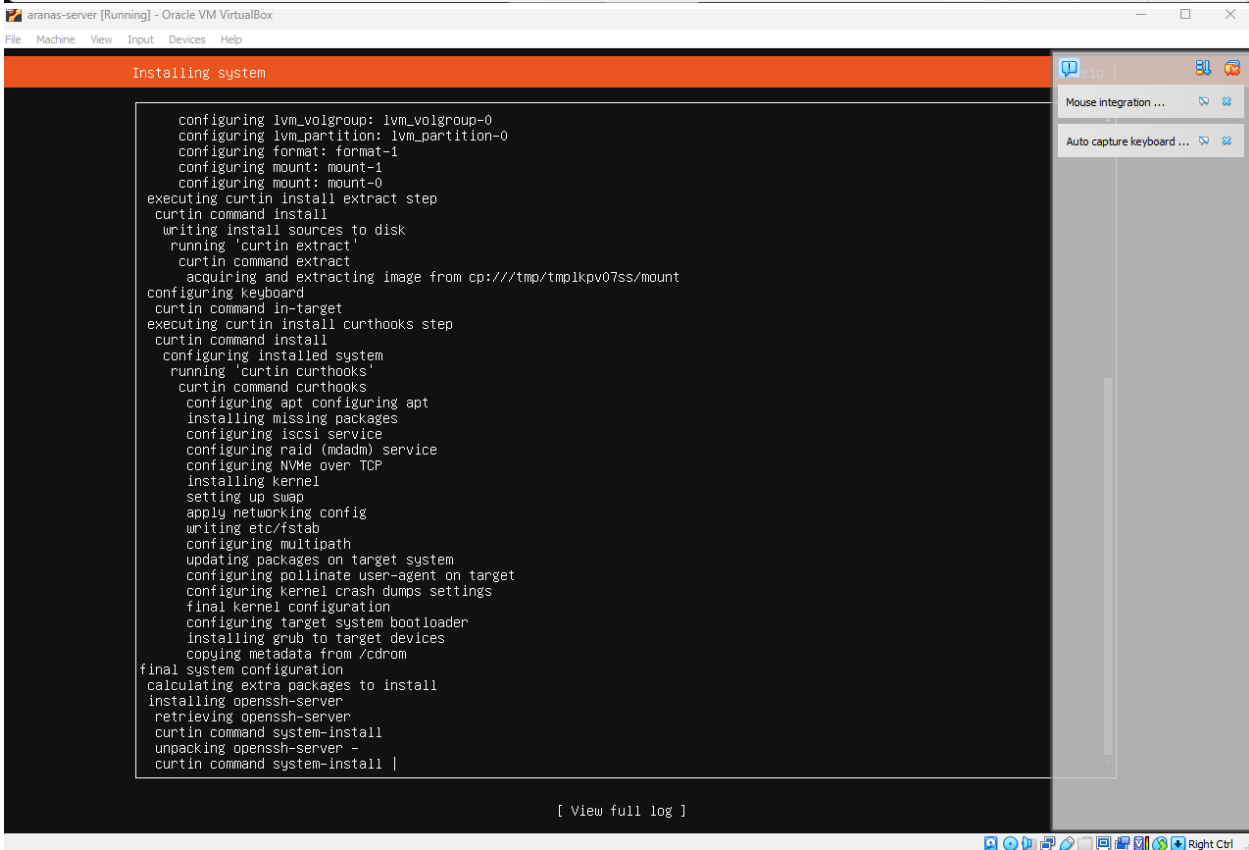
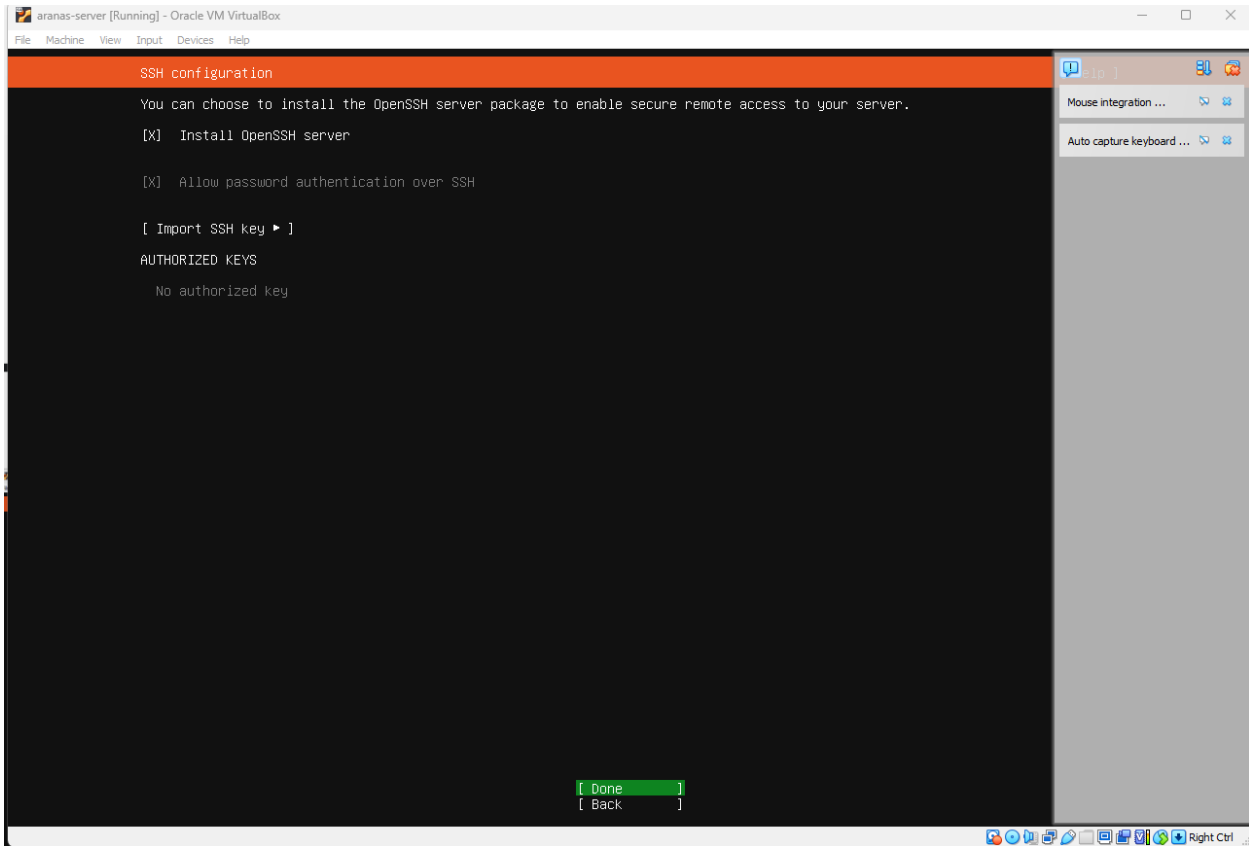


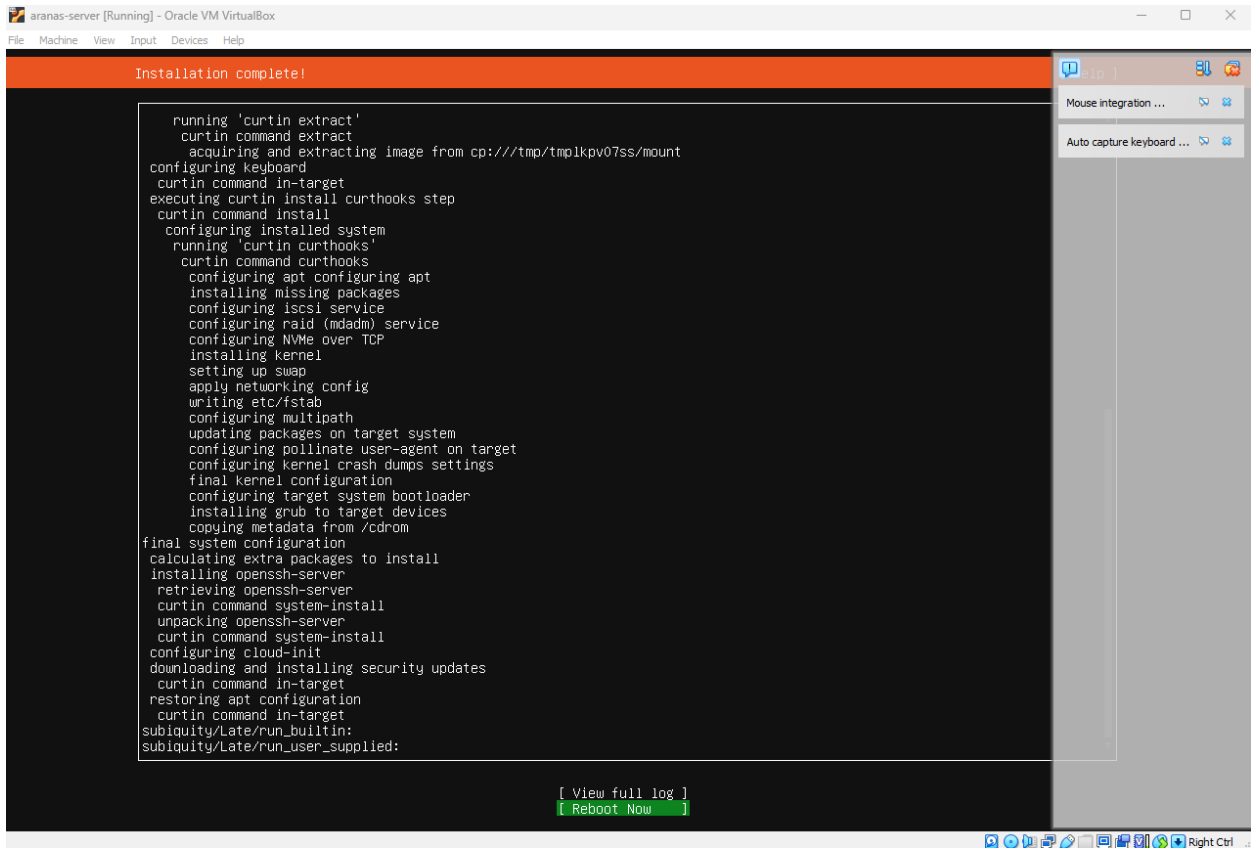




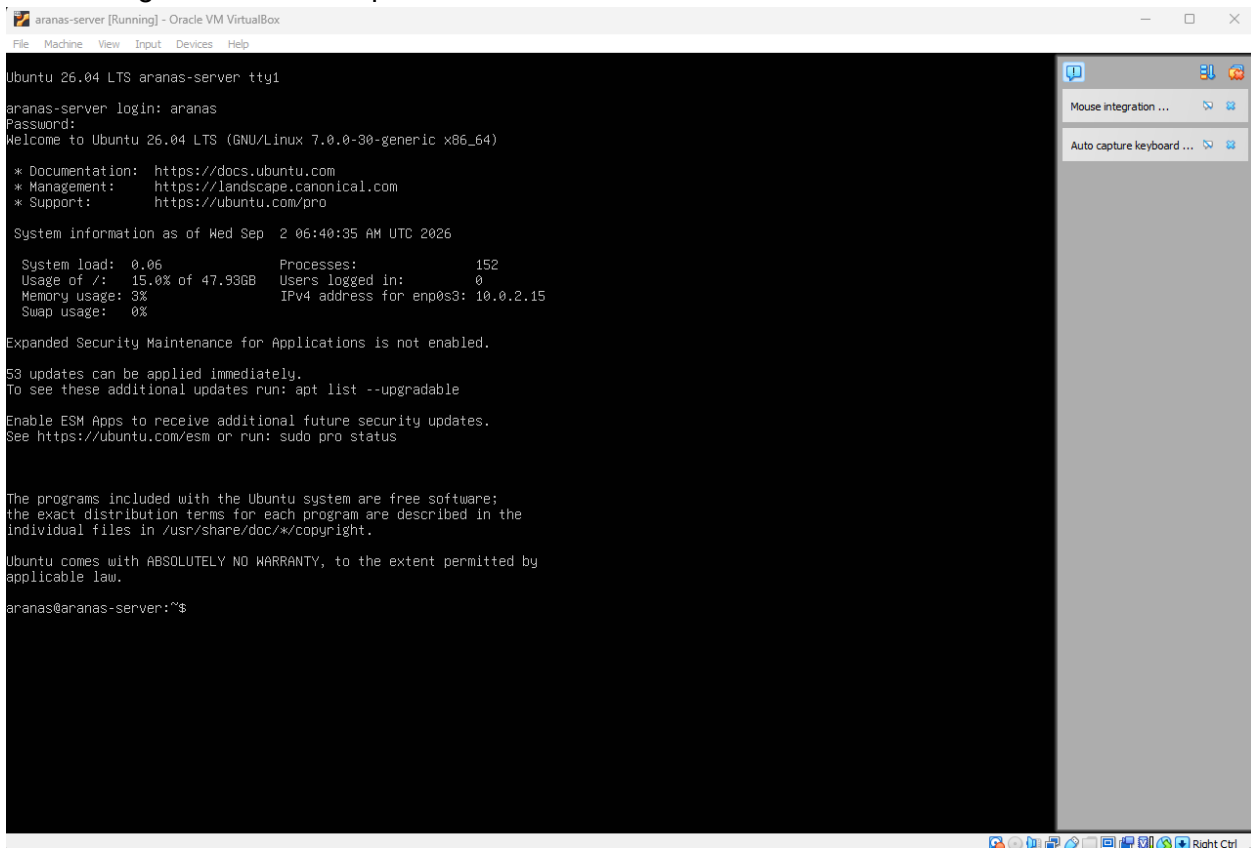








Connecting server to desktop



```
System load: 0.06      Processes:      152
Usage of /: 15.0% of 47.93GB Users logged in: 0
Memory usage: 3%      IPv4 address for enp0s3: 10.0.2.15
Swap usage: 0%
```

Expanded Security Maintenance for Applications is not enabled.

63 updates can be applied immediately.

To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```
aranas@aranas-server:~$ sudo systemctl status ssh
```

```
[sudo: authenticate] Password:
```

```
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Wed 2026-09-02 06:37:41 UTC; 5min ago
   Invocation: 69eb9820a1bd4d15bb4c79f00d2766db
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 5564 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5567 (sshd)
     Tasks: 1 (limit: 8592)
   Memory: 1.6M (peak: 2.5M)
     CPU: 21ms
   CGroup: /system.slice/ssh.service
           └─5567 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

```
sep 02 06:37:41 aranas-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
```

```
sep 02 06:37:41 aranas-server sshd[5567]: Server listening on 0.0.0.0 port 22.
```

```
sep 02 06:37:41 aranas-server sshd[5567]: Server listening on :: port 22.
```

```
sep 02 06:37:41 aranas-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

```
aranas@aranas-server:~$ sudo apt update
```

```
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
```

```
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
```

```
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
```

```
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
```

```
65 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
aranas@aranas-server:~$
```

See <https://ubuntu.com/esm> or run: `sudo pro status`

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aranas@aranas-server:~\$ sudo systemctl status ssh

[sudo: authenticate] Password:

```
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   Active: active (running) since Wed 2026-09-02 06:37:41 UTC; 5min ago
     Invocation: 69eb9820a1bd4d15bb4c79f00d2766db
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
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```

Sep 02 06:37:41 aranas-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...

Sep 02 06:37:41 aranas-server sshd[5567]: Server listening on 0.0.0.0 port 22.

Sep 02 06:37:41 aranas-server sshd[5567]: Server listening on :: port 22.

Sep 02 06:37:41 aranas-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

aranas@aranas-server:~\$ sudo apt update

Hit:1 <http://security.ubuntu.com/ubuntu> InRelease

Hit:2 <http://ph.archive.ubuntu.com/ubuntu> InRelease

Hit:3 <http://ph.archive.ubuntu.com/ubuntu> InRelease

Hit:4 <http://ph.archive.ubuntu.com/ubuntu> InRelease

55 packages can be upgraded. Run 'apt list --upgradable' to see them.

aranas@aranas-server:~\$ sudo apt install openssh-server-y

Error: Unable to locate package openssh-server-y

aranas@aranas-server:~\$ sudo apt install openssh-server -y

openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).

The following packages were automatically installed and are no longer required:

```
linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generi
linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
```

Use 'sudo apt autoremove' to remove them.

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55

aranas@aranas-server:~\$

the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

```
aranas@aranas-server:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Wed 2026-09-02 06:37:41 UTC; 5min ago
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Sep 02 06:37:41 aranas-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...

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Sep 02 06:37:41 aranas-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

aranas@aranas-server:~\$ sudo apt update

Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease

Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease

Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease

Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease

55 packages can be upgraded. Run 'apt list --upgradable' to see them.

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linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14

Use 'sudo apt autoremove' to remove them.

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55

aranas@aranas-server:~\$ sudo systemctl enable ssh

Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.

Executing: /usr/lib/systemd/systemd-sysv-install enable ssh

Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.

Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.

aranas@aranas-server:~\$

aranas-server [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
Process: 5564 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 5567 (sshd)
  Tasks: 1 (limit: 8592)
Memory: 1.6M (peak: 2.5M)
   CPU: 21ms
  CGroup: /system.slice/ssh.service
          └─5567 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Sep 02 06:37:41 aranas-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...

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aranas@aranas-server:~\$ sudo apt update

Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease

Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease

Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease

Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease

55 packages can be upgraded. Run 'apt list --upgradable' to see them.

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linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic

linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14

Use 'sudo apt autoremove' to remove them.

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55

aranas@aranas-server:~\$ sudo systemctl enable ssh

Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.

Executing: /usr/lib/systemd/systemd-sysv-install enable ssh

Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.

Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.

aranas@aranas-server:~\$ sudo systemctl start ssh

aranas@aranas-server:~\$ ip a

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:24:df:cb brd ff:ff:ff:ff:ff:ff
    altname enx08002724dfcb
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85734sec preferred_lft 85734sec
    inet6 fe80::a00:27ff:fe24:dfcb/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
```

aranas@aranas-server:~\$

Right Ctrl

Desktop connected to server.

